A thermoacoustic combined cooling, heating, and power (CCHP) system for waste heat and LNG cold energy recovery

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Abstract

Waste energy recovery is crucial for reducing greenhouse gas emissions and alleviating the increasing demand for central power grids. In the paper, theoretical analyses have been carried out on a novel thermoacoustic combined cooling, heating, and power system for simultaneously recovering waste heat and liquefied natural gas cold energy. The system is designed based on acoustic impedance matching between the thermoacoustic and alternator units for maximizing the overall exergy efficiency. The efficiency, cost savings, and emission reductions of the optimal system are investigated. With cold and waste heat temperatures of 130 K and 500 K, the system achieves an overall exergy efficiency of 24.1%, allowing 78.4 MWh of primary fuel energy reduction, and correspondingly, 30.6 tons of CO_2 emissions reduction per year. Investigations are further performed on the system parametric sensitivities. A method for adjusting the alternator external electric impedance is proposed to allow for matching piston displacement within design parameters. Besides, comparisons are made with alternative thermoacoustic cogeneration models. The results show that the proposed system can operate effectively for various supply modes, which is promising for adaptation in different applications. Finally, comparisons are drawn with similar existing technologies and future developments are discussed.

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Keywords

Combined cooling, heating, and power (CCHP); Waste heat; Cold recovery; Thermoacoustic; Stirling; Combined heating and power (CHP)

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Nomenclature

Symbols

a	speed of acoustic, m/s
A_p	area of the piston, m^2
C_e	electrical capacitance, m^3/Pa
C_s	energy cost saving, €
COP_c	coefficient of performance of the electrical chiller
c_{el}	price of electricity, €/kWh
c_{ng}	price of natural gas, €/kWh
D_p	diameter of the piston, m
d_h	hydraulic diameter, m
e	mass-specific total gas energy, J/kg
f	frequency, Hz
f_{el}	CO_2 emission factor of electricity, $kgCO_2$/kWh
f_{ng}	CO_2 emission factor of natural gas, $kgCO_2$/kWh
f_D	Darcy friction factor
F	viscous pressure gradient
I_1	electrical current, A
K	total spring stiffness, N/m
K_m	mechanical spring stiffness, N/m
K_g	gas spring stiffness, N/m
k_g	gas thermal conductivity, W/(m·K)
L_e	electrical inductance, H
M	moving mass, kg
M_{opt}	optimal moving mass, kg
Nu	Nusselt number
N_k	enhanced gas conduction coefficient
P_1	pressure amplitude, Pa
P_m	mean pressure, Pa
Pr	Prandtl number
Pe	Peclet number
Q_c	cooling power, W
Q_a	heat transferred from/to ambient, W
Q_h	pumped heat power, W
Q_L	LNG cold energy, W
Q_w	waste heat energy, W
Q_{low}	cold energy from a cold source, W
Q_{high}	heat energy from a heat source, W
q	instantaneous axial heat flux, W/m^2
Q_f	film heat transfer, W
R_a	real parts of acoustic impedance, Pa·s /m^3
R_m	mechanical damping coefficient, N·s/m
$R_{e,in}$	internal electrical resistances, Ω
$R_{e,out}$	external electrical resistances, Ω
Re	Reynolds number
r_h	hydraulic radius, m
S_x	wetted perimeter, m
X_a	imaginary parts of acoustic impedance, Pa·s /m^3
X_e	electrical reactance, Ω
r_{amp}	power amplification factor
T	temperature, K
T_0	ambient temperature, K
T_c	cooling temperature, K
T_h	heating temperature, K
T_L	temperature of liquefied natural gas, K
T_w	temperature of the waste heat, K
T_{low}	low temperature of a cold source, K
T_{high}	high temperature of a heat source, K
U_1	volume flow rate, m^3/s
v	complex piston velocity, m/s
V_b	volume of back space, m^3
W	acoustic power, W
W_l	acoustic power to load, W
$W_{R,i}$	acoustic power at the inlet of the thermoacoustic regenerator, W
$W_{R,o}$	acoustic power at the outlet of the thermoacoustic regenerator, W
$W_{eng,i}$	acoustic power at the inlet of the thermoacoustic engine, W
$W_{eng,o}$	acoustic power at the outlet of the thermoacoustic engine, W

W_e	electric power, W
S_p	piston displacement amplitude, m
Z_a	acoustic impedance, Pa·s /m^3
$Z_{a,s}$	acoustic impedance at one side of the double-sided alternator, Pa·s /m^3

Greek letters

$\eta_{ex,eng}$	exergy efficiency of the engine
$\eta_{ex,ove}$	exergy efficiency of overall system
$\eta_{en,ove}$	energy efficiency of overall system
η_{alt}	acoustic-to-electric efficiency of the alternator
$\eta_{alt,opt}$	optimal alternator efficiency
$\eta_{ex,the}$	exergy efficiency of the thermoacoustic unit
η_{con}	efficiency of conventional power plant
η_{boi}	efficiency of boiler
ϕ	porosity
θ	phase difference, rad
τ	transduction coefficient, N/A
τ_{anu}	annual operating time, hour
ω	angular frequency, rad/s
γ	specific heat ratio
ε	average height of surface irregularities

Special symbol

i	notation for an imaginary value
Re []	Real part of
| |	magnitude of the complex number
*	complex conjugation

Abbreviations

A	ambient-temperature heat exchanger
CCP	combined cooling and power
CHP	combined heating and power
CCHP	combined cooling, heating, and power
ER	emission reduction
FES	fuel energy saving
H	high-temperature heat exchanger
L	low-temperature heat exchanger
LNG	liquefied natural gas
R	regenerator
RT	resonance tube
T	thermal buffer tube

1 Introduction

World energy demands are expected to dramatically increase by up to 28% from 2015 to 2040 [1] with growing populations and economies. Technologies that recover waste energy, i.e., waste heat and waste cold energy, provide the opportunity to improve energy utilization efficiency and reduce greenhouse gas emission. Over recent decades, the world has focused on recovering waste heat energy from renewable energy sources (solar and geothermal) [2,3] as well as industrial processes [4,5]. Current active technologies for utilizing waste heat include sorption systems [6], mechanically driven heat pumps [7] and organic Rankine cycles [8]. A number of studies have recently focused on waste cold energy recovery [9,10], given the potential of liquefied natural gas (LNG), since LNG releases around 830 kJ/kg of cold energy during the regasification process before combustion. Given the global annual LNG regasification capacity of 824 MTPA [11], the potential cold energy corresponds to 6.838×10^{14} kJ per year. However, such a substantial amount of available cold energy remains underutilized in the traditional regasification stage.

The combination of waste heat recovery and LNG available energy utilization can further improve energy utilization efficiency, given the higher potential availability between the two states. However, most current systems are based on the Rankine cycle, which is cost-effective only for energy utilization on a scale of megawatts. In addition to waste energy existing on a large scale, there is abundantly available waste energy on a much smaller scale in distributed energy systems. Therefore, the development of technically and economically viable energy solutions for recovering such small-scale waste energy remains a challenge.

Thermoacoustic technology may offer an alternative for harnessing such small-scale waste energy. This heat-driven technology can be applied for heating [12], cooling [13,14] or power generation [15,16], and works by converting thermal energy into acoustic power in the thermoacoustic engine and using acoustic power to pump heat via a thermoacoustic heat pump or to generate electricity via an alternator. Thermoacoustic technology involves no harmful ozone-depleting gases or mechanical moving parts and offers potential low-cost installation based on simple components. Thermoacoustic systems can effectively operate at power ranges from a few tens of watts to dozens of kilowatts. Thermoacoustic technology, therefore, represents a promising environmentally friendly, reliable, and simple way of recovering small-scale waste energy.

Thermoacoustic systems for recovering waste heat energy have been developed by many researchers. In 2010, de Blok developed a looped thermoacoustic refrigerator [17], and the system achieved a cooling power of 78.2 W and an exergy efficiency of 9.3% at heating and cooling temperatures of 442 K and 240 K, respectively. In 2012,

Yu *et al.* developed a prototype of a thermoacoustic electricity generator for the utilization of low-grade heat energy [18], and the experimental results depicted an onset temperature difference of about 120 K. In 2014, Mumith *et al* designed a thermoacoustic engine for waste heat recovery in food manufacturing [19]. The calculation results show the engine can deliver 1 kW of output acoustic power with an exergy efficiency of about 17%. In 2017, Wang *et al.* numerically studied a four-stage looped thermoacoustic Stirling power generator [20]. The simulation result depicted a maximum electric power of 1.2 kW and the highest exergy efficiency of around 20%. In 2018, Jin *et al.* conducted experimental studies of a looped three-stage thermoacoustic electric generator [21], and the demonstrator achieved an exergy efficiency of 5% at the heating temperature of 367 K. Later, Hamood *et al.* developed a prototype of a two-stage thermoacoustic electricity generator to recover waste heat from exhaust gases in internal combustion engines [22]. The demonstrator exhibited electric power of around 50 W with an exergy efficiency of 11% at a heating temperature of 590 K. In 2020, Yang *et al.* theoretically studied a traveling-wave phase-change thermoacoustic engine [23]. When using the engine to drive an alternator, an exergy efficiency of 27 % and electrical power of 10 W were expected to be achieved at a heating temperature of 340 K. Recently, Xu *et al.* proposed a thermoacoustic refrigerator with hybrid gas-liquid resonators for waste heat recovery [24]. At heating and cooling temperatures of 400 K and 270 K respectively, their calculation results demonstrated that a four-stage cooling system was able to provide a cooling power of 1.7 kW and an exergy efficiency of around 22 %.

In addition to using thermoacoustic systems for waste heat utilization, other efforts have been directed at recovering the available cold energy from LNG regasification processes. Wang *et al.* proposed a thermoacoustic Stirling generator for LNG cold and waste heat recovery. Their system numerically achieved an output electric power of 2.3 kW and an exergy efficiency of about 25 % at heating and cooling temperatures of 500 K and 110 K respectively [25]. Hou *et al.* numerically studied a thermoacoustic Stirling electrical generator for LNG cold energy recovery. With cooling and heating temperatures of 110 K and 303 K respectively, their system exhibited the highest exergy efficiency of about 21 % [26].

However, current investigations of thermoacoustic systems for waste energy recovery have been limited to single-generation applications (i.e., either for cooling, heating, or electricity), which may hinder their wider usage in a variety of applications. Another limitation of the current waste-recovery thermoacoustic system is the low power generation from either waste heat or available cold temperature gradients. In this paper, we proposed a first-ever concept of thermoacoustic system to tackle these two key issues: (1) multiple thermodynamic cycles are integrated in one system to enable different energy-conversion processes, and hence resulting in combined provisions of

heating, cooling and electricity; (2) combined utilizations of waste heat and cold energy allows for the larger power amplification factor relatively to single-temperature gradient thermoacoustic systems (demonstrated in Section 3), leading to significantly higher power per unit volume of heat exchanger. To the best of the authors' knowledge, no research has been done on combining a thermoacoustic multi-generation system with the utilization of both waste heat and LNG cold energy.

This paper aims to design a thermoacoustic multi-generation system for waste heat and LNG cold energy recovery based on a maximum exergy-based objective, and to present the working characteristics of the system in detail. In Section 2, system configuration and performance indicators are introduced. In Section 3, the performance of the thermoacoustic engine units when driven by three types of energy sources (heat, cold energy, and combined heat and cold) are discussed. Section 4 presents the design method of the thermoacoustic trigeneration system. Section 5 shows the working characteristics of the thermoacoustic trigeneration system, including system performance, effects of key operation conditions, and parameter sensitivity control. In Section 6, further investigations are conducted to explore thermoacoustic cogeneration systems, and comparisons are made to the thermoacoustic trigeneration system. In Section 7, the proposed thermoacoustic multi-generation systems are further compared to the existing technologies and their future potential applications are introduced. Section 8 presents conclusions.

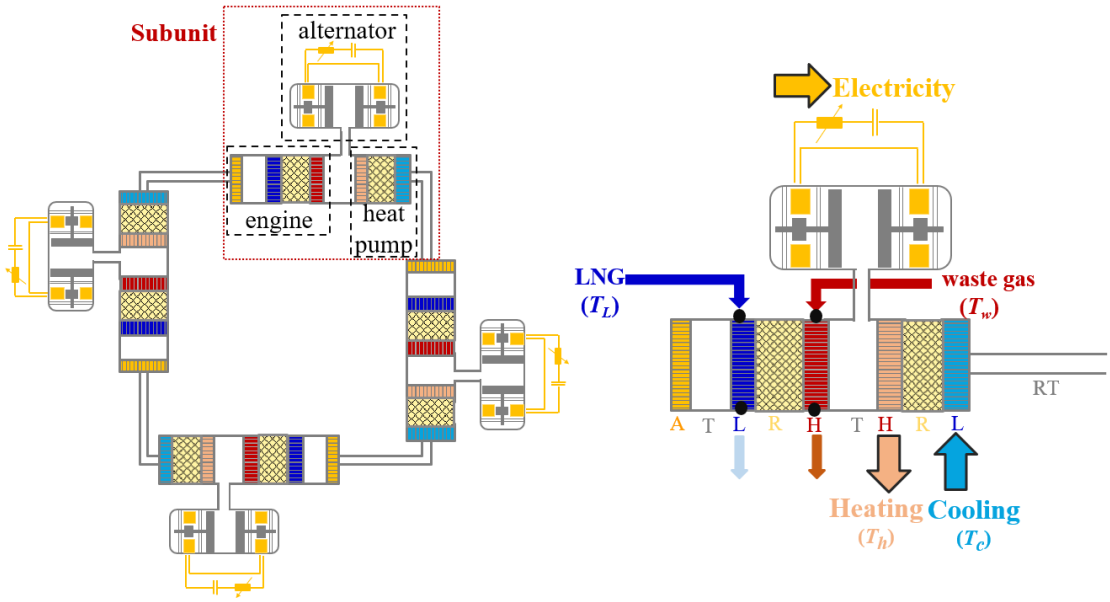
2. System description

2.1 Configuration

Figure 1(a) shows a schematic of the four-stage thermoacoustic trigeneration system. The system incorporates four identical energy-conversion subunits, as shown in Figure 1(b), in a loop through resonance tubes. In each subunit, a thermoacoustic engine unit is directly connected to a thermoacoustic heat pump, between which a double-sided linear alternator is coupled as a branch. The thermoacoustic engine unit consists of an ambient-temperature heat exchanger, a thermal buffer tube, a low-temperature heat exchanger, a regenerator, and a high-temperature heat exchanger. The engine unit converts thermal energy into acoustic power when a temperature gradient is applied along the regenerator. Waste heat flows through the high-temperature heat exchanger, and LNG flows through the low-temperature heat exchanger. The heat pump unit consists of a high-temperature heat exchanger, a regenerator, and a low-temperature heat exchanger; the heat pump unit uses the acoustic power generated to pump heat from the cold to the ambient side, thus generating a temperature gradient along with the regenerator. The alternator consists of two identical resonant linear motor units in a balanced opposed arrangement with clearance seal pistons, which

compress a gas in a common front chamber and are all enclosed in a bolted pressure vessel (further details to be shown in Figure 6). The coils of the linear alternators are connected in series with a variable electric impedance, thereby dissipating the generated electricity.

The working principles of the overall system are as follows: when the axial temperature gradient across the regenerator of the engine unit exceeds a critical value, a self-excited thermoacoustic oscillation is generated and acoustic power is amplified. Part of the amplified acoustic power is used in the alternator unit to generate electric power. The remaining acoustic power is consumed by the heat pump unit which pumps heat from the low-temperature to high-temperature exchangers, where cooling and heating power are exchanged, respectively. The remaining acoustic power at the outlet of the heat pump is transmitted by a resonance tube to the next subunit. Finally, acoustic power is recycled into the next subunit.



(a) (b)

Figure 1. Schematic of the thermoacoustic trigeneration system for waste heat and LNG cold energy recovery: (a) overall four-stage system; (b) enlarged view of an energy-conversion subunit which is highlighted in the red rectangle in Fig. 1a. A is the ambient-temperature heat exchanger, T is the thermal buffer tube, L is the low-temperature heat exchanger, R is the regenerator, H is the high-temperature heat exchanger and RT is the resonance tube. The different colors in the heat exchangers and regenerators represent different temperatures.

2.2 Performance indicators

Both energy and exergy efficiencies are of interest in understanding system performance. The energy efficiency is evaluated as useful cooling, heating, and work output per unit heat input of the cycle:

$$\eta_{en,ove} = \frac{Q_c + Q_h + W_e}{Q_w} \tag{1}$$

where Q_w, Q_h, and Q_c are the waste heat thermal energy, pumped heat and cooling power, respectively, and W_e is the output electric power.

The energy efficiency is often insufficient to evaluate system performance in that it cannot quantify the performance of a system compared to a reversible cycle. Further, energy efficiency only takes into account the quantity of the thermodynamic losses rather than the quality of the energy. The exergy efficiency is used to assess the effectiveness of energy resource utilization in a thermal system. In the present case, the exergy efficiency of the overall system, $\eta_{ex,ove}$, is the ratio of the sum of the exergy flow output from the CCHP system to the exergy flow-rate in the heat and cold supply stream:

$$\eta_{ex,ove} = \frac{Q_c(\frac{T_0}{T_c} - 1) + Q_h(1 - \frac{T_0}{T_h}) + W_e}{Q_L(\frac{T_0}{T_L} - 1) + Q_w(1 - \frac{T_0}{T_w})} \tag{2}$$

where Q_L is the supplied LNG cold energy. T_0, T_c, T_h, T_L, and T_w are the ambient temperature, cooling temperature, heating temperature, LNG temperature and waste heat temperature, respectively.

In addition to energy and exergy performance, the fuel energy saving (FES) is also of interest, as the total energy saved in the proposed CCHP system in comparison with separate dedicated conventional energy systems consisting of a boiler, electrical chiller, and conventional power stations. The annual fuel energy saving can be written as:

$$FES = (\frac{W_e}{\eta_{con}} + \frac{Q_h}{\eta_{boi}} + \frac{Q_c}{COP_c})\tau_{anu} \tag{3}$$

where η_{con}, η_{boi} and COP_c are the efficiency of the conventional power plant, efficiency of the boiler, and coefficient of performance of the electrical chiller, respectively. η_{con}, η_{boi} and COP_c for the conventional energy systems are taken here to be 40% [27], 82% [28] and 2.37 [29] respectively. The annual operating time, τ_{anu}, is 7200 hours.

The annual energy cost saving of the proposed system, C_s, can be estimated by:

$$C_s = (\frac{W_e}{\eta_{con}} c_{el} + \frac{Q_h}{\eta_{boi}} c_{ng} + \frac{Q_c}{COP_c} c_{el})\tau_{anu} \tag{4}$$

where c_{el} is the price of electricity, 0.174 €/kWh in the UK; c_{ng} is the price of natural gas, 0.058 €/kWh in the UK [30].

The potential CO_2 emission reduction (ER) has been the main driving force for the increasing interest in the market growth of waste energy utilization technologies. The annual reduction of CO_2 emission by using the proposed system can be estimated as:

$$ER = (\frac{W_e}{\eta_{con}} f_{el} + \frac{Q_h}{\eta_{boi}} f_{ng} + \frac{Q_c}{COP_c} f_{el})\tau_{anu} \tag{5}$$

where f_{el} is the CO_2 emission factor of natural gas, 0.5246 $kgCO_2/kWh$; f_{ng} is the CO_2 emission factor of natural gas, 0.1836 $kgCO_2/kWh$ [30].

3. Comparison of thermoacoustic engine units driven by different energy sources

The capacity of the thermoacoustic engine to generate acoustic power from an established temperature gradient makes it possible to exploit different types of energy sources. Before any investigations take place on an overall trigeneration system, the influence of energy sources on the thermoacoustic engine unit, needs to be compared first. In this section, therefore, we explore the performance of the thermoacoustic engine units when driven by three types of energy sources: heat, cold energy, and combined heat and cold, at respective different heating and cooling temperatures (T_{high} and T_{low}), as illustrated in Figure 2. Each thermoacoustic engine unit is sandwiched by two pistons that act as acoustic impedance boundaries. To ensure a fair comparison for different energy sources, the acoustic impedances at the boundaries are optimized to achieve the highest efficiency in each calculation. The engine exergy efficiency and power amplification factor are expressed as

$$\eta_{ex,eng} = \frac{W_{eng,o} - W_{eng,i}}{Q_{low}(\frac{T_0}{T_{low}} - 1) + Q_{high}(1 - \frac{T_0}{T_{high}})} \tag{6}$$

$$r_{amp} = \frac{W_{R,o}}{W_{R,i}} \tag{7}$$

where $W_{eng,i}$ and $W_{eng,o}$ are the acoustic power at the inlet and outlet of the thermoacoustic engine, respectively. $W_{R,i}$ and $W_{R,o}$ are the acoustic power at the inlet and outlet of the engine regenerator, respectively. Q_{high} is the heat energy from the heat source, Q_{low} is the cold energy from the cold source. T_{high} and T_{low} are the temperatures of the heat and cold sources, respectively. r_{amp} is the power amplification factor for the input acoustic power.

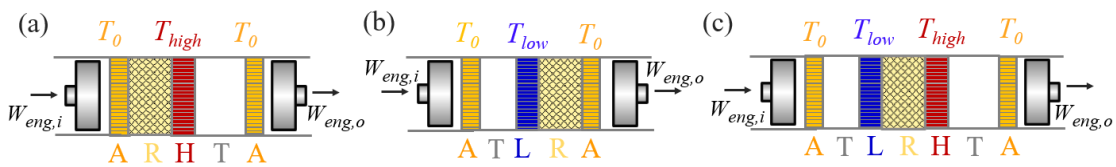
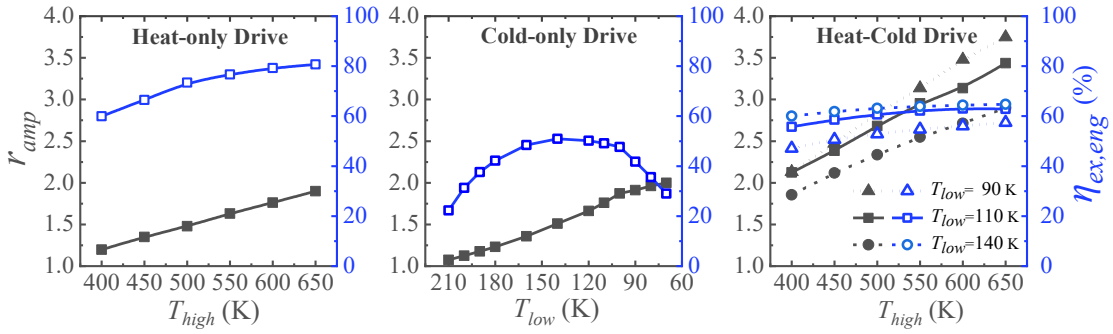


Figure 2. Thermoacoustic engine units driven by different energy sources from (a) heat energy (b) cold energy (c) combined heat and cold energy.

Calculations are implemented using the thermoacoustic code Sage [31] (see detailed description in Section 5.1). Figure 3 shows the performance of thermoacoustic engine units when using different types of energy sources at respective different temperatures. A decrease in the cold source temperature leads to a significant increase in the power amplification factor but a slight decrease in engine efficiency. Compared to single-source driven engines, the proposed combined-source engine achieves a much higher power amplification factor with useful efficiencies, and thus is favorable for increasing the system power capacity.



(a) (b) (c)

Figure 3. Simulated performance of the thermoacoustic engine units at different heating and cooling temperatures when driven by different energy sources of (a) heat energy, (b) cold energy, and (c) combined heat and cold energy. P_m is 4 MPa, T_0 is 293 K, and $W_{en,i}$ is 1 kW at f of 55 Hz.

4. Design of the thermoacoustic trigeneration system

Section 3 shows that the thermoacoustic engine driven by the combined heat and cold energy achieves a much higher power amplification factor than a similar system driven by a single energy source. Following this encouraging result, this section further presents the design and optimization of the thermoacoustic trigeneration system driven by the combined waste heat and LNG cold energy. Section 4.1 introduces the design method, followed by the detailed analyses of the output characteristics of the thermoacoustic unit and the coupling of the alternator in Section 4.2 and 4.3. In Section 4.4, based on the designed alternator, further investigations are carried out on the effect of the external electric impedance to give a better understanding of the alternator characteristics.

4.1 Design method

In the thermoacoustic CCHP system, the thermoacoustic unit (which consists of an engine, a heat pump, and a resonance tube) largely depends on the acoustic impedance of the alternators, which acts as the acoustic load to the thermoacoustic unit.

In addition, the required acoustic impedance for coupling the thermoacoustic unit also affects alternator efficiency. Therefore, the acoustic impedance coupling between the thermoacoustic unit and the alternator is crucial to the overall performance. In such a strongly coupled system, the joint optimization of the alternator with the thermoacoustic unit as a whole system means that each unit may not work at its respective optimal performance. Besides, due to a large number of control parameters, it is challenging to realize the optimal impedance match between the thermoacoustic unit and alternator if a complete model of the whole system is used. An effective way of solving this problem is to introduce a load acoustic impedance to decouple the thermoacoustic and alternator units. Given the objective of maximizing the overall exergy efficiency, firstly it is required to find the optimal load acoustic impedance for the thermoacoustic unit to ensure the highest efficiency of the thermoacoustic unit; then the alternator should be designed to achieve the required load acoustic impedance as well as the maximum alternator efficiency simultaneously. In this way, they can be both operated at the optimal state, so the coupled system is able to achieve the highest efficiency at its full potential.

The design procedure for the present work is illustrated in Figure 4. The characteristics of the required load acoustic impedances Z_a for the thermoacoustic unit are firstly analyzed independently; the optimal decision variable Z_a is obtained when maximizing the thermoaocustic unit efficiency η_{the}. Second, the relationship between the alternator efficiency and the load acoustic impedances is analysed; the optimal Z_a is taken into the above relationship, and the alternator piston diameter D_p and moving mass M are optimized to achieve the highest efficiency of the alternator η_{alt}. Finally, the thermoacoustic unit and the alternator are optimally coupled in terms of highest overall efficiency when they can both reach their respective highest efficiency. The operation constraints for system design are given in Table 1.

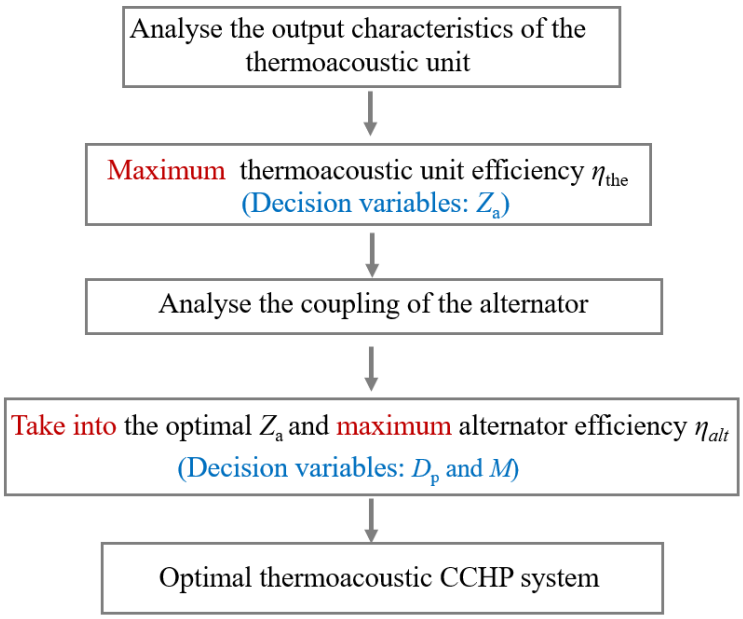


Figure 4. Procedure for design of the thermoacoustic CCHP system

Table 1. Operation constraints for the design of the system

Variables	Definition	Values
P_m	mean pressure, MPa	4
T_0	ambient temperature, K	293
T_w	waste heat temperature, K	500
T_L	liquefied natural gas temperature, K	130
T_h	heating temperature, K	323
T_c	cooling temperature, K	283

4.2 Output characteristics of the thermoacoustic unit

Analogous to electric impedance, acoustic impedance is a convenient property for characterizing effects that occur when the sound wave meets the boundary between two phases. Here, a load acoustic impedance, defined as the ratio of pressure wave P_1 to volume flow rate U_1, is used to replace the alternator. i.e.

$$Z_a = \frac{P_1}{U_1} = R_a + iX_a = \frac{|P_1|}{|U_1|}\cos\theta_{p-U} + i\frac{|P_1|}{|U_1|}\sin\theta_{p-U} \tag{8}$$

where R_a and X_a are the real and imaginary parts of acoustic impedance, respectively; $|\,|$ is the amplitude of the complex number, and $\theta_{p\text{-}U}$ is the phase difference between the pressure and volume flow rate waves.

The Sage program is used to calculate system performance under different load acoustic impedances, to explore the optimal impedance in terms of highest efficiency

for the thermoacoustic unit. The dimensions of the thermoacoustic unit are presented in Table 2. The exergy efficiency of the thermoacoustic unit is defined as

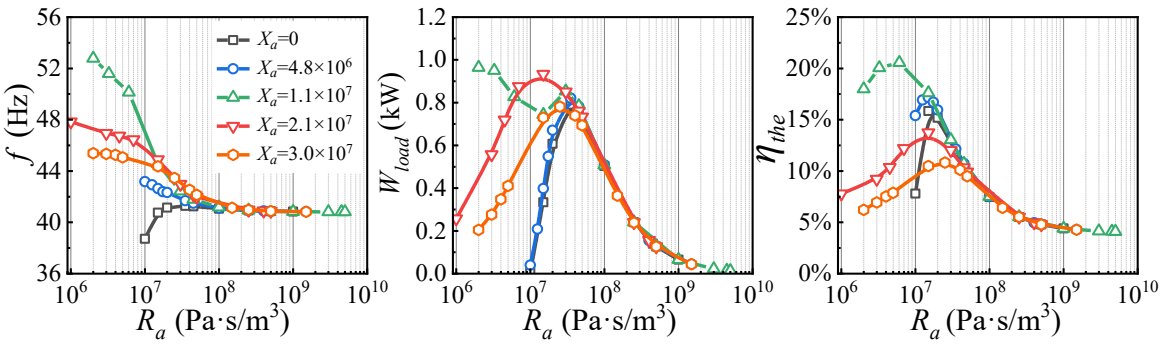
$$\eta_{ex,the} = \frac{Q_c(\frac{T_0}{T_c}-1)+Q_h(1-\frac{T_0}{T_h})+W_l}{Q_L(\frac{T_0}{T_L}-1)+Q_w(1-\frac{T_0}{T_w})} \tag{9}$$

where W_l represents the acoustic power input to the load.

Table 2. Dimensions of the thermoacoustic unit. A, L, R and H are shown in Figure 1. All dimensions are in millimeters (mm).

Unit	Part	Diameter	Length	Others
engine	A	100	25	shell-tube type, 35% in porosity, 5 in tube diameter
	L		30	shell-tube type, 30% in porosity, 1 in tube diameter
	R		40	88% in porosity, 0.018 in wire diameter
	H		30	shell-tube type, 30% in porosity, 1 in tube diameter
heat pump	H	81	26	shell-tube type,40% in porosity, 5 in tube diameter
	R		20	90% in porosity, 0.027 in wire diameter
	L		27	shell-tube type,40% in porosity, 1 in tube diameter
resonance tube	-	17	2900	-

Figure 5 shows the effect of the load acoustic impedance on the frequency, output acoustic power to the load, and exergy efficiency of the thermoacoustic unit. Based on the calculation results, the optimal load acoustic impedance Z_a is $(7+11i)\times10^6$ Pa·s/m^3 for the highest efficiency of the thermoacoustic unit. Using this value, the frequency is 48.5 Hz, the highest exergy efficiency of the thermoacoustic unit is 20.5 % and the output acoustic power to the load is 823 W.



(a) (b) (c)

Figure 5. Effect of load acoustic impedance (unit of X_a is Pa·s/m^3) on the (a) frequency, (b) output acoustic power to the load, and (c) exergy efficiency for the thermoacoustic unit. P_m is 4 MPa, T_0 is 293 K, T_L is 110 K, T_w is 450 K, T_h is 323 K and T_c is 283 K.

4.3 Coupling of the alternators

The relationship among the alternator efficiency, the alternator parameters and the load acoustic impedance is analysed. Figure 6 shows the key parameters of the alternator. The governing equations for an alternator are as follows [32],

$$\frac{\tau}{A_p}U_1=(R_{e,in}+R_{e,out}+iX_e)I_1 \tag{10}$$

$$\frac{\tau}{A_p}I_1+\frac{R_m+i(\omega M-K/\omega)}{A_p^{\,2}}U_1=P_1 \tag{11}$$

where P_1 is the pressure wave at the inlet of the alternator, I_1 is the electrical current of the alternator, U_1 is the volume flow rate associated with the alternator's motion, A_p is the area of the piston, R_m is the mechanical damping, τ is the transduction coefficient between acoustic power and electric power, and X_e is the electrical reactance.

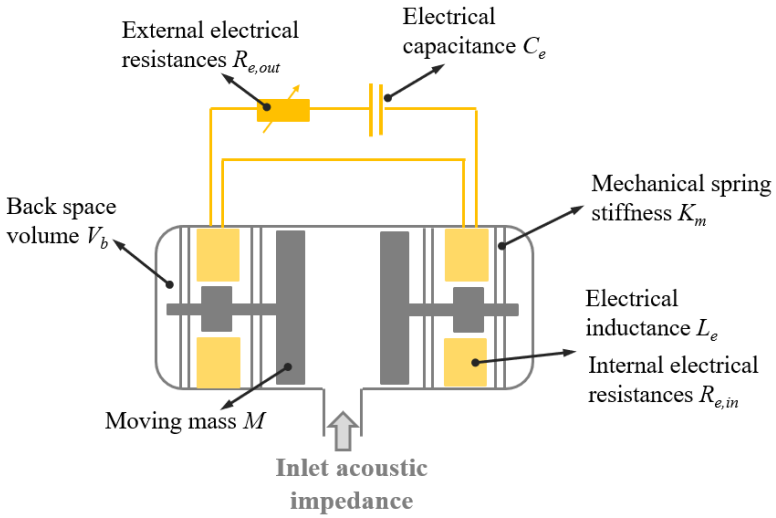


Figure 6. Elements of the alternator

The total spring stiffness K consists of mechanical spring stiffness K_m and gas spring stiffness K_g that is obtained through

$$K_g=\frac{\gamma P_m A_p}{V_b} \tag{12}$$

where γ is the specific heat ratio, P_m is the mean pressure, and V_b is the volume of the alternator back space. The acoustic impedance at the inlet of the alternator is

$$R_a+iX_a=\frac{P_1}{U_1} \tag{13}$$

By solving Eqs. (10), (11) and (13), the ratio I_1/U_1 can be expressed in terms of the acoustic impedance:

$$\frac{I_1}{U_1}=\frac{A_p{}^2R_a-R_m+i(A_p{}^2X_a-(\omega M-K/\omega))}{A_p\tau} \tag{14}$$

The acoustic power input to the alternator, W_a, and the electric power produced by the alternator, W_e, can be calculated respectively by

$$W_a=\frac{1}{2}\mathrm{Re}[P_1U_1^*]=\frac{\tau}{2A_p}\mathrm{Re}[U_1^*I_1]+\frac{R_m}{2A_p{}^2}|U_1|^2 \tag{15}$$

$$W_e=\frac{1}{2}\mathrm{Re}[(I_1R_{e,out})I_1^*]=\frac{\tau}{2A_p}\mathrm{Re}[U_1I_1^*]-\frac{1}{2}R_{e,in}|I_1|^2 \tag{16}$$

where Re [] refers to the real part of a complex number, and the superscript * indicates complex conjugation. In Eq. (15), the term $(R_m/2A_p{}^2)\,|U_1|^2$ of the input acoustic power is consumed by the mechanical resistance, and the remainder of the input acoustic power, $\tau\mathrm{Re}\,[U_1^*I_1]\,/(2A_p)$, is converted to electric power. In Eq. (16), the electric power $\tau\mathrm{Re}[U_1I_1^*]/(2A_p)$, is partly dissipated by the electric resistance via a portion of $R_{e,in}|I_1|^2/2$, and the remainder is delivered as the output electric power W_e. Taking the ratio of Eqs. (15), (16) and substituting Eq. (14) into the equations gives the acoustic-to-electric efficiency of the alternator, η_{alt},

$$\eta_{alt}=\frac{W_e}{W_a}=1-\frac{R_m}{R_aA_p{}^2}-\frac{R_{e,in}A_p{}^2}{R_a\tau^2}[(\frac{R_m}{A_p{}^2}-R_a)^2+(\frac{\omega M-K/\omega}{A_p{}^2}-X_a)^2] \tag{17}$$

where $(\omega M\text{-}K/\omega)/A_p{}^2\text{-}X_a=0$, η_{alt} achieves a maximum value if all the other variables are constant. Under these conditions, the optimal moving mass M_{opt} and the Eq. (17) can be correspondingly rewritten as

$$M_{opt}=K/\omega^2+X_aA_p{}^2/\omega \tag{18}$$

$$\eta_{alt,opt}=1-\frac{R_m}{A_p{}^2R_a}-\frac{R_{e,in}A_p{}^2}{R_a\tau^2}(\frac{R_m}{A_p{}^2}-R_a)^2 \tag{19}$$

In the optimization design, the acoustic impedance at the inlet of the alternator must be equal to the optimal load acoustic impedance, i.e. $(7+11i)\times10^6$ Pa·s/m^3, which is obtained in Section 4.2. For a double-sided alternator, the inlet acoustic impedance at one side of the alternator, $Z_{a,s}$, is twice that of the total acoustic impedance. The value of $Z_{a,s}$, therefore, is taken as $(1.4+2.2i)\times10^7$ Pa·s/m^3. When incorporating the optimal load impedance into Eqs. (18) and (19) and keeping the other parameters in Table 3 constant, Figure 7 shows the effect of piston diameter on the optimal moving mass and alternator efficiency. Though increasing piston diameter results in higher alternator

efficiency when the diameter is below 90 mm, the corresponding increase in the moving mass and piston diameter may also lead to large vibration intensity. To achieve a compromise, D_p is chosen to be 70 mm, with M_{opt} of 2.84 kg and $\eta_{alt,opt}$ of 85%. Table 3 lists all the parameters of the alternator.

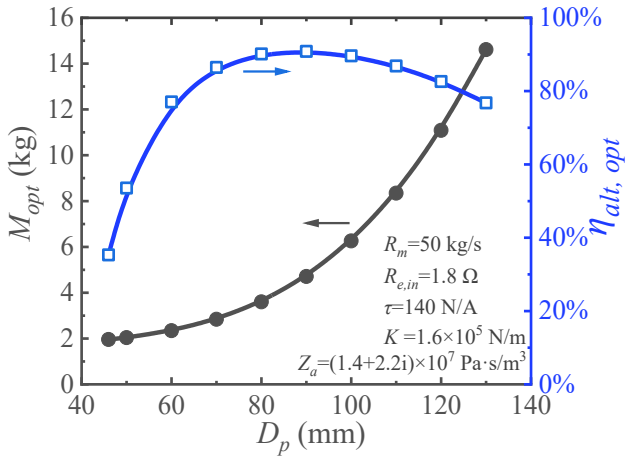


Figure 7. Effect of the piston diameter on the optimal moving mass and optimal alternator efficiency.

Table 3. Designed parameters for the alternator in one side

D_p (mm)	M (kg)	R_m (kg/s)	$R_{e,in}$(Ω)	L_e (H)	τ (N/A)	K_m (N/m)	V_b (L)
70	2.84	25	1.8	0.155	140	1.4x10^5	1.5

4.4 Effect of external electric impedance on the designed alternator

To further understand the characteristic of the designed alternator above, the effect of the external electric impedance associated with electric resistances $R_{e,out}$ and capacitance C_e is investigated. The alternator inlet acoustic impedance and efficiency can be expressed as [33]:

$$Z_a = \frac{1}{A_p^{\,2}}[R_m + i(\omega M - \frac{K}{\omega}) + \frac{\tau^2}{R_{e,in} + R_{e,out} + iX_e}] \tag{20}$$

$$\eta_{alt} = \frac{\tau^2 R_{e,out}}{\tau^2 (R_{e,out} + R_{e,in}) + R_m((R_{e,out} + R_{e,in})^2 + X_e^{\,2})} \tag{21}$$

If no external electric capacitance C_e is connected in the electric circuit, X_e is equal to ωL_e; if the external electric capacitance C_e is connected in the electric circuit, X_e is equal to ωL_e-1/(ωC_e). Figure 8 shows the effect of the external electric impedance on the designed alternator with or without external electric capacitance C_e. Where C_e is not connected, the alternator efficiency exceeds 85% when the $R_{e,out}$ is within a range of 40 to 100 Ω; where C_e is connected, a similar high efficiency is achieved when C_e is larger

than 35 μF and $R_{e,out}$ is smaller than 110 Ω. The optimal acoustic impedance of (1.4+2.2i) ×10^7 Pa·s/m^3, is achieved when C_e is not connected and with $R_{e,out}$ of 100 Ω.

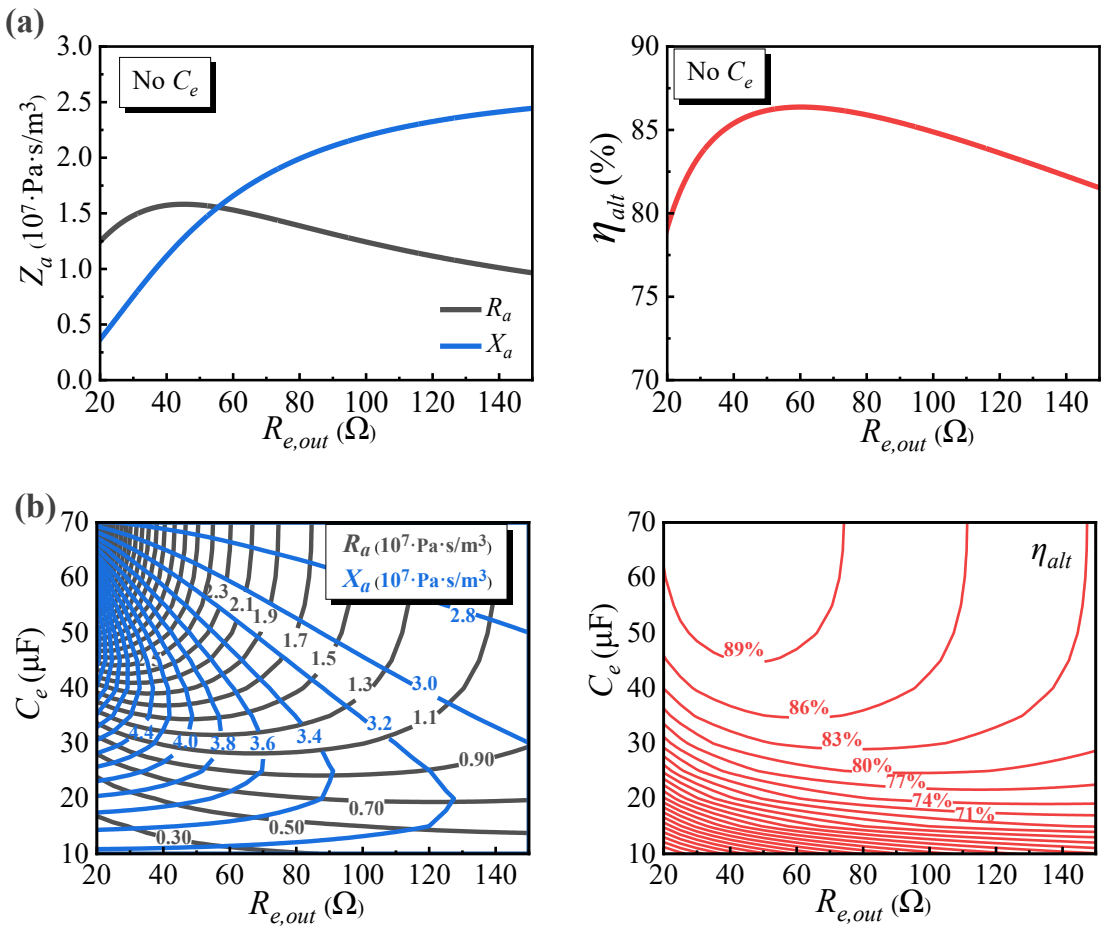


Figure 8. Effects of external electric impedance on inlet acoustic impedance and efficiency in the designed alternator: (a) when no external electric capacitance C_e is connected; (b) when the electric external electric capacitance C_e is connected. The other dimensions of the alternator are listed in Table 3.

5. Working characteristics of the thermoacoustic trigeneration system

5.1 Simulation model

Based on the optimized dimensions from Section 4, the working characteristics of the thermoacoustic trigeneration system are investigated in detail by using the Sage program, which has been used for thermoacoustic refrigerators [34,35], thermoacoustic electric generators [36,37], pulse tube coolers [38,39] and Stirling devices [40,41].The program consists of many model instances such as regenerators, heat exchangers, pistons, tubes and so on, each of which is a collection of component building blocks with parameter input values. The program solves the equations for conservation of mass, momentum, and energy for each model instance. The governing equations of continuity, momentum, and energy equations for the gas domain in the model are [42]

$$\frac{\partial \rho A}{\partial t}+\frac{\partial \rho u A}{\partial x}=0 \tag{22}$$

$$\frac{\partial \rho u A}{\partial t}+\frac{\partial u \rho u A}{\partial x}+\frac{\partial P_1}{\partial x}A-FA=0 \tag{23}$$

$$\frac{\partial \rho e A}{\partial t}+P_1\frac{\partial A}{\partial t}+\frac{\partial}{\partial x}(u\rho e A+uP_1A+q)-Q_w=0 \tag{24}$$

where P_1, u, A and ρ are the pressure wave, mean-flow velocity, cross section area and gas density, respectively. The variable e is mass-specific total gas energy with $e=\kappa+u^2/2$ (κ is mass-specific internal gas energy). The factor F represents the viscous pressure gradient, Q_f is the heat flow per unit length due to heat transfer, and q is the conductive heat flow in the axial direction. Terms F, Q_f and q are formulated as [42]

$$F=-(f_D/d_h+K_{los}/l)\rho u|u|/2 \tag{25}$$

$$Q_f=Nu(k/d_h)S_x(T_w-T) \tag{26}$$

$$q=-N_k k_g\frac{\partial T}{\partial x}A \tag{27}$$

where f_D, d_h and K_{los} are the Darcy friction factor, hydraulic diameter and total local loss coefficient respectively. Nu, k_g, S_x and (T_w-T) are the Nusselt number, gas conductivity, wetted perimeter, and temperature difference between the negative z surface and section-average, respectively. N_k is an axial-conductivity enhancement ratio that is represented as a ratio of the effective gas conductivity to the molecular conductivity. Table 4 lists empirical terms f_D, Nu and N_k used in the simulation model for the present work.

Table 4. Empirical terms of Darcy friction factor f_D, Nusselt number Nu, *and* axial-conductivity enhancement ratio N_k in the simulation model [42] for the present work. *Re*, *Pr* and *Pe* are the Reynolds number, Prandtl number and Peclet number, respectively. Other symbols are defined in the nomenclature section.

Component	f_D	Nu	N_k
Heat exchanger	Laminar: $f_D=\frac{64}{Re}$ Turbulent: $f_D=0.11(\frac{\varepsilon}{d_h}+\frac{68}{Re})^{0.25}$	Laminar steady: $Nu=6$ Turbulent: $Nu=0.036Re^{0.8}(\frac{l}{d_h})^{-0.055}Pr^{0.33}$	Laminar: $N_k=1$ Turbulent:

			$N_k = 0.022Re^{0.75}Pr$
Regenerator	$f_D = \frac{129}{Re} + 2.91Re^{-0.103}$	$Nu = (1 + 0.99Pe^{0.66})\phi^{1.79}$	$N_k = 1 + 0.5Pe^{0.66}\phi^{-2.91}$
Cylinder	$f_D = 0$	$\mathrm{Re}(Nu) = \begin{cases} \sqrt{2V_aPr} \text{ if } R_t \le 7.7\sqrt{2V_aPr} \\ \frac{0.13R_t}{\mathrm{In}(\frac{0.35R_t}{\sqrt{2V_aPr}})} \text{ otherwise} \end{cases}$ $\mathrm{Im}(Nu) = \frac{\tanh(\frac{0.4}{a})}{1+3a}\mathrm{Real}(Nu)$ where $a = \sqrt{\frac{R_t}{7.8V_aPr}}$	$N_k = 1 + 0.014R_t$
Tube	Laminar: $f_D = \frac{64}{Re}$ Turbulent: $f_D = 0.11(\frac{\varepsilon}{d_h} + \frac{68}{Re})^{0.25}$	Laminar steady: $Nu = 6$ Turbulent: $Nu = 0.036Re^{0.8}(\frac{l}{d_h})^{-0.055}Pr^{0.33}$	Laminar: $N_k = 1$ Turbulent: $N_k = 0.022Re^{0.75}Pr$

Previous experimental validation of the models in the program showed reasonable accuracy against experimental data. For example, in a cascade two-stage thermoacoustic cooler by Xu *et al.* [39], the deviations between simulations and experiments for the cooling power were in the range of 4 to 12%. Likewise, deviations from the model efficiencies in Guo *et al* 's work on a Stirling refrigerator were within 10% [43]. The discrepancies between the simulation model and experiments mainly arise from the limited heat transfer model due to the one-dimensional formulation and the underestimated loss from acoustic streaming and static heat losses [44]. Nevertheless, the work described above show that the simulation results are within engineering accuracy.

5.2 Efficiency, cost savings, and emission reductions of the system

A summary of the efficiency, energy cost savings, and emission reductions of the thermoacoustic trigeneration system for waste heat and LNG cold energy recovery is shown in Table 5, using the optimal dimensions defined in section 4. The results indicate that this system can convert LNG cold energy of 6.16 kW at 130 K and waste heat of 8.26 kW at 500 K to produce a combined heating power of 3.55 kW, a cooling power of 2.19 kW and an electric power of 2.27 kW, with an overall exergy efficiency of 24.1% and an energy efficiency of 96.9%. Our cost savings and emissions assessments indicate that, this system has the potential to save around 78480 kWh of

primary fossil-fuel energy consumption and to displace around 30.6 tons of CO_2 emission per year.

We note that the economic assessment here is limited. A comprehensive analysis of an energy system would require evaluation of annualized investment costs, including installation, operation, and maintenance costs, to show the payback time and the levelized cost of energy for a hypothetical thermoacoustic CCHP system relatively to a conventional system. However, this is not currently possible owing to significant uncertainties for these costs, since thermoacoustic technology has not been available in the market and most components are customized.

Table 5. Energy performance, energy cost savings and emission reductions of the thermoacoustic trigeneration system by using waste heat and LNG cold energy.

Operating conditions		
P_m	mean pressure, MPa	4
T_0	ambient temperature, K	293
T_w	waste heat temperature, K	500
T_L	liquefied natural gas temperature, K	130
T_h	heating temperature, K	323
T_c	cooling temperature, K	283
$R_{e,out}$	external electrical resistances, Ω	100
Energy performance		
f	frequency, Hz	48.1
Q_L	liquefied natural gas cold energy, kW	6.16
Q_w	waste heat energy, kW	8.26
Q_h	pumped heat power, kW	3.55
Q_c	cooling power, kW	2.19
W_e	electrical power, kW	2.27
$\eta_{en,ove}$	energy efficiency of the overall system,%	96.9
$\eta_{ex,ove}$	exergy efficiency of the overall system,%	24.1
Energy cost savings and emission reductions		
FES	annual fuel energy saving, MWh/year	78.4
C_s	annual energy cost saving, €/year	10075
ER	annual potential CO_2 emission reduction, tCO_2/year	30.6

5.3 Axial distribution of phase fields

The key to achieving efficient energy conversion in a thermoacoustic system is to establish traveling-wave acoustic fields, i.e., the in-phase relationship between P_1 and

U_1, across the regenerators [45]. Figure 9 shows the axial distribution of the phase difference between the pressure wave and volume flow rate in the thermoacoustic engine and heat pump. The *x*-axis begins at the inlet of the engine unit and ends at the outlet of the heat pump unit. The results show that the traveling-wave acoustic fields dominate engine regenerator (phase difference from -9.3° to 28.7°) as well as the heat pump regenerator (phase difference from 10.7° to 15.8°), which result in high acoustic power and efficient energy conversion provided by each thermoacoustic unit.

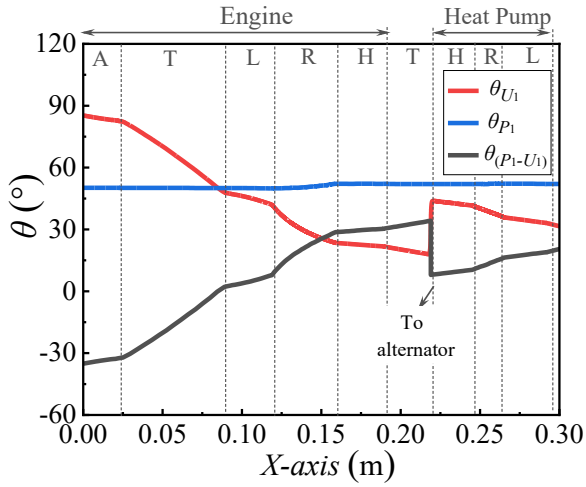


Figure 9. Axial distribution of phase difference between the pressure wave and volume flow rate in the thermoacoustic engine and heat pump. A is the ambient-temperature heat exchanger, T is the thermal buffer tube, L is the low-temperature heat exchanger, R is the regenerator, and H is the high-temperature heat exchanger.

5.4 The influence of key working parameters

In this section, the influences of key working parameters – the external electric impedances, LNG and waste heat temperatures, cooling and heating temperatures – on the performance of the thermoacoustic trigeneration system are presented. The parametric sensitivity analysis aims to find out how the system performs by varying the key working parameters, which provides a reference for implementing the thermoacoustic trigeneration system in various applications.

5.4.1 Effects of external electric impedances

The external electric impedance has a significant impact on alternator inlet acoustic impedance and is thus crucial to overall system performance. Figure 10 shows the calculation results of system performance with different external electric resistance and capacitance. The result shows that a balance exists between the power capacity and exergy efficiency for different external electric impedance values: when C_e equals 20 μF, the system achieves a maximum Q_c of 10.9 kW and Q_h of 19.2 kW with a low η_{ove} of 8.9 %; the reverse performance applies when there is no C_e, such that the maximum

η_{ove} of 17.6 % is achieved with a low Q_c of 2.43 kW, Q_h of 3.76 kW and W_e of 2.44 kW. In a heat-driven thermoacoustic system, the operating frequency is the fundamental frequency of self-excited oscillation. The results show that the external electric resistance slightly influences the frequency.

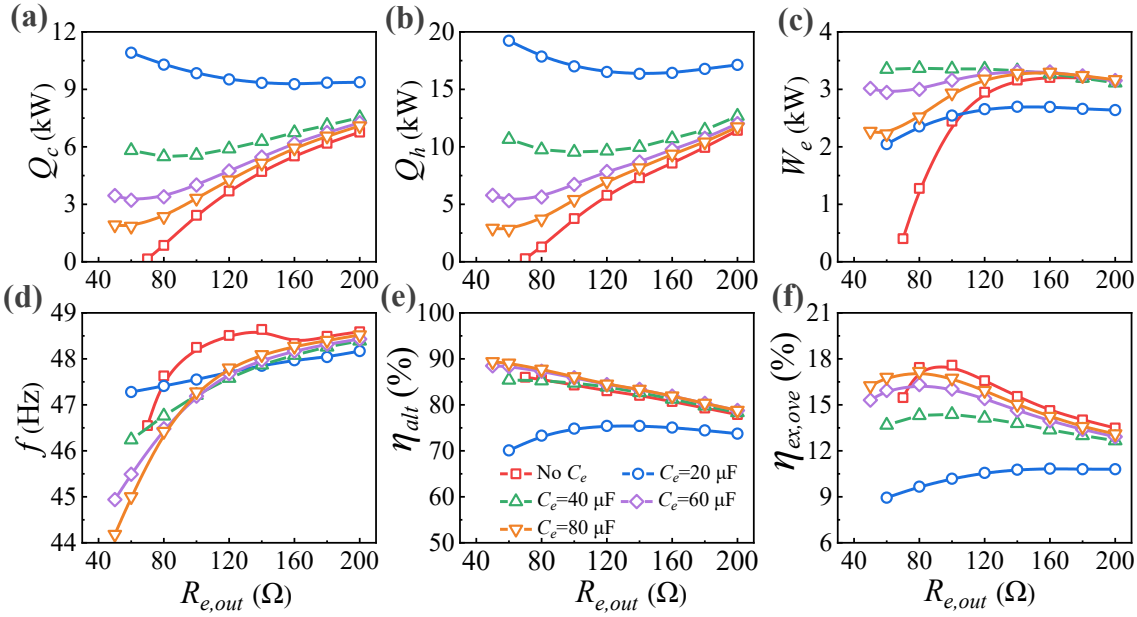


Figure 10. Dependence of external electric resistance $R_{e,\,out}$ and capacitance C_e on the (a) cooling power Q_c, (b) heating power Q_h, (c) electric power W_e, (d) frequency f, (e) alternator efficiency η_{alt} and (f) overall exergy efficiency η_{ove}, T_w is 450 K, T_L is 110 K, T_h is 323 K, T_c is 283 K, T_0 is 293 K, and P_m is 4 MPa.

5.4.2 Effects of LNG and waste heat temperatures

In practice, LNG and waste heat temperatures may vary for various conditions. This is because LNG is generally stored in insulated storage tanks during transportation, where its liquefaction temperature is influenced by the filling pressure of the tank. With a common tank pressure from 0.3 to 10 bar, the LNG temperature usually varies from 110 K to 170 K. Furthermore, the temperature of the available waste heat also varies depending on the industrial sector and the process, within a majority range of 323 K to 773 K. Figure 11 shows the effects of LNG and waste heat temperatures on system performance. An increase in waste heat temperature has a favorable effect on the power capacity. Larger LNG temperatures dramatically decrease the power capacity while contributing to larger efficiency. As T_L increases from 110 K to 170 K, the cooling power, heating power, and electric power are decreased by a factor of 3.4, 6.9 and 3.1 respectively while the exergy efficiency is increased by a factor of 1.85, at T_w of 600 K. The results also show the LNG and waste heat temperatures only slightly affect frequency and alternator efficiency. Based on Eqs. (20) and (17), this is because temperature change only results in a small change in acoustic impedance (e.g., for T_w

from 400 K to 650 K, Z_a only changes from (1.41+2.09i) ×10^7 to (1.37+1.81i) ×10^7 Pa·s/m^3), so the frequency and alternator efficiency only vary marginally. For a typical T_w of 500 K and T_L of 110 K, the system achieves a combined cooling power of 3.81 kW at 283 K, a heating power of 7.5 kW at 323 K and an electric power of 3.78 kW, with an overall exergy efficiency of 18.5 %.

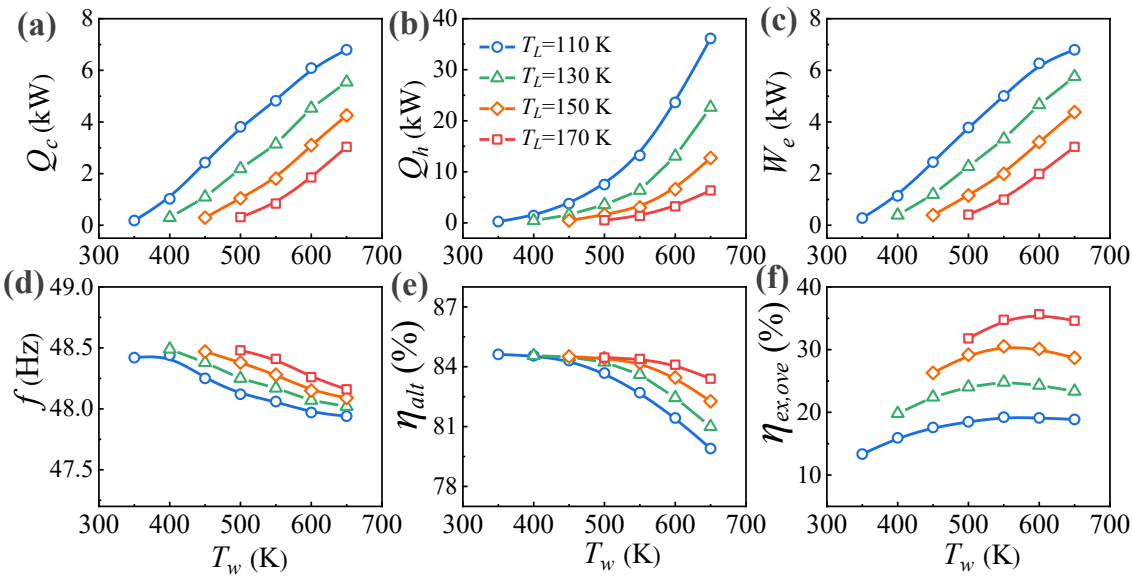


Figure 11. Dependence of waste heat temperature T_w and LNG temperature T_L on the (a) cooling power Q_c, (b) heating power Q_h, (c) electric power W_e, (d) frequency f, (e) alternator efficiency η_{alt} and (f) overall exergy efficiency η_{ove}, for T_h is 323 K, T_c is 283 K, T_0 is 293 K, P_m is 4 MPa, R_e is 100 Ω, and no C_e is connected.

5.4.3 Effects of cooling and heating temperatures

The operating cooling temperature varies for different applications: below 255 K for a freezer, between 273 and 277 K for a fridge, from 270 to 290 K for a wine cooler and between 293 to 300 K for an air conditioner. Similarly, the heating temperature varies for different heating applications such as space heating, domestic hot water or industrial processes. Figure 12 shows the effects of cooling and heating temperatures on system performance. Higher cooling temperatures are favorable for increasing power capacity due to the smaller temperature gradient across the heat pump regenerator. A similar trend is also observed by decreasing heating temperatures. The operating temperatures negligibly impact on acoustic impedance, which further leads to the independence of the frequency and alternator efficiency. For the given range of parameters, the thermoacoustic trigeneration system displays high overall exergy efficiency. This means the system can effectively operate at different working temperatures for various applications.

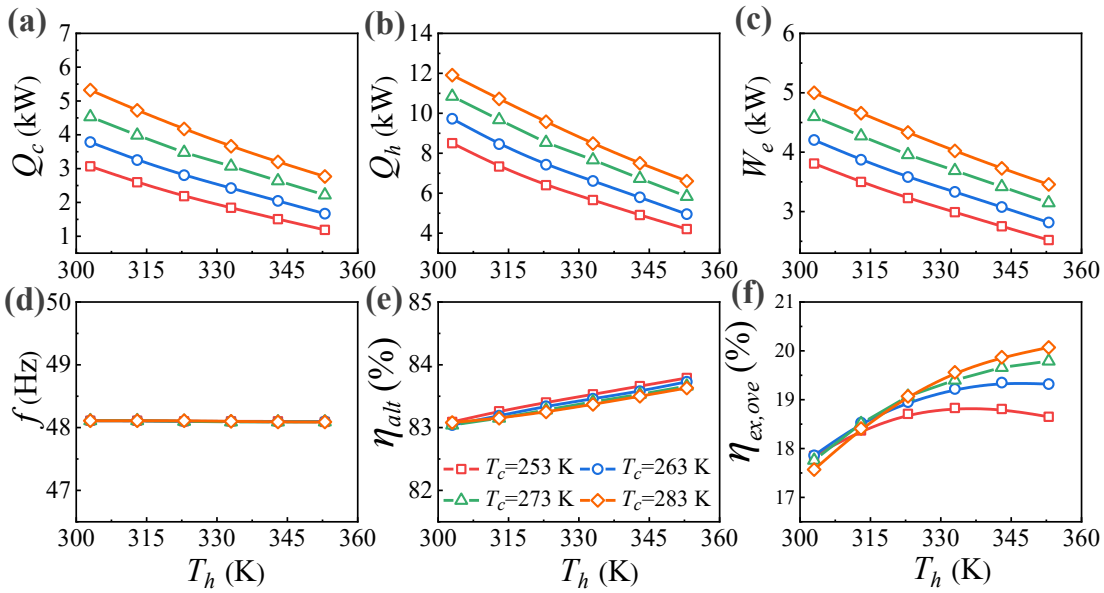


Figure 12. Dependence of the (a) cooling power Q_c, (b) heating power Q_h, (c) electric power W_e, (d) frequency f, (e) alternator efficiency η_{alt} and (f) overall exergy efficiency η_{ove} in relation to the heating temperature T_h and cooling temperature T_c, for T_w is 523 K, T_L is 110 K, T_0 is 293 K, P_m is 4 MPa, R_e is 100 Ω, and when no C_e is connected.

5.5 Parameter sensitivity control

In practice, it is inevitable for the operating parameters to deviate from the design, which can upset the desired energy balance in the system. These deviations can be explained by combining Eqs. (10), (11) and (13) to obtain the following two equations,

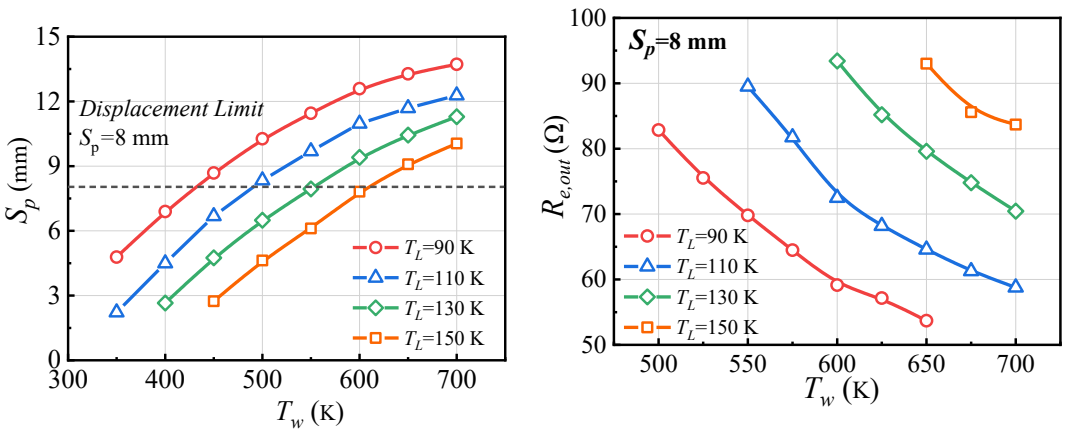
$$R_a A_p^{\,2} = R_m + \frac{(R_{e,out} + R_{e,in})\tau^2}{[(R_{e,out} + R_{e,in})^2 + X_e^{\,2}]} \tag{28}$$

$$X_a A_p^{\,2} = \omega M - \frac{K}{\omega} - \frac{X_e \tau^2}{[(R_{e,out} + R_{e,in})^2 + X_e^{\,2}]} \tag{29}$$

The multiplication of Eq. (28) by $|v|^2/2$ (where v is a complex expression of piston velocity) shows the energy balance of the alternator: the term $R_a|A_p v|^2/2$ on the left represents the input acoustic power into the alternator, the first term on the right, $R_m|v|^2/2$, denotes the dissipated acoustic power by the mechanical resistance of the piston, and second term on the right characterizes the acoustic power transduced into electric power. This means that, if the parameters deviate from the designed value, the frequency and piston displacement change and achieve a new balance, which may lead to problems if the piston displacement unexpectedly exceeds the designed value. However, for most existing heat-driven thermoacoustic systems, it is difficult to actively modulate the thermal parameters within very short time due to significant thermal response hysteresis in the heat exchangers and regenerators. A feasible solution

is to adjust the external electric resistance to keep the displacement of the piston constant at the allowable value [46].

In a practical system, the waste heat and LNG temperatures may fluctuate. We therefore firstly explore how the piston displacement changes according to different temperatures, and then discuss how to keep the piston displacement constant at the allowable value by varying the external electric resistance to avoid it exceeding the limitation. Figure 13(a) shows the influence of the LNG and waste heat temperatures on the piston displacement S_p. The simulation results demonstrate that the larger waste heat temperatures and smaller LNG temperatures call for a larger piston displacement. By actively adjusting the external electric resistance, the piston displacement could be kept constant at a typical allowable value of 8 mm which depends on the size of the piston, as shown in Figure 13(b). The result show that a system with a larger waste heat temperature or smaller LNG temperature, which leads to a larger displacement, can be accommodated by using a smaller value of external electric resistance to limit the piston displacement. In other words, the required external electric resistance should be increased at a smaller waste heat temperature or larger LNG temperature. For example, at the largest waste heat temperature and the smallest LNG temperature, the required external electric resistance for the constant displacement is 52 Ω; while at the smallest waste heat temperature and the largest LNG temperature, the required external electric resistance is increased to 94 Ω.

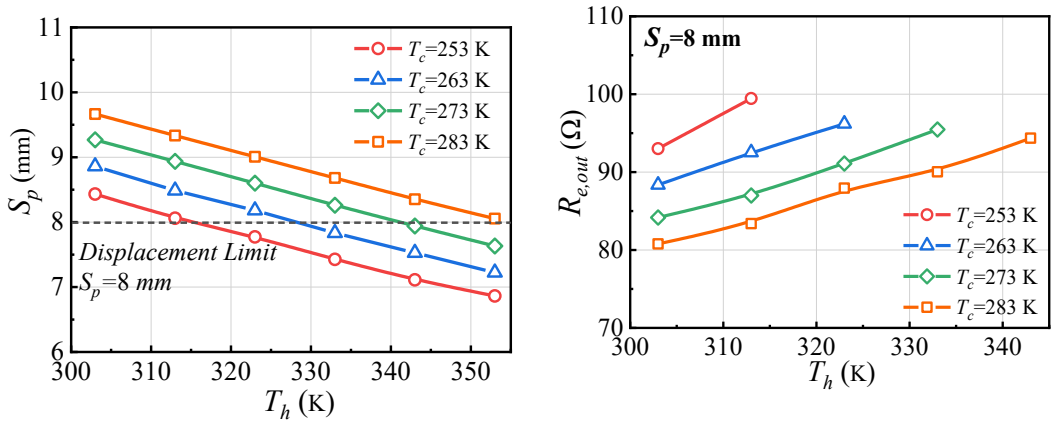


(a) (b)

Figure 13. (a) Effects of LNG and waste heat temperatures on the piston displacement S_p, (b) the required external electric resistance $R_{e,out}$ to keep piston displacement at a typical allowable value of 8 mm.

Similar calculations for different cooling and heating temperatures are depicted in Figure 14. Figure 14(a) shows that the piston displacement decreases with the heating temperatures, and increases with increased cooling temperatures. For a cooling

temperature of 263 K as an example, the piston displacement will exceed the limitation of 8 mm if the heating temperature is lower than 330 K. This can also be avoided by adjusting the external electric resistance to keep the piston displacement constant at the allowable value. Figure 14(b) shows the required external electric resistance for keeping the displacement constant at a typical allowable value of 8 mm. The results show that a lower electric resistance is required to limit the displacement at a lower heating temperatures and higher cooling temperatures. In other words, the required external electric resistance must be increased at a higher heating temperatures and lower cooling temperatures.



(a) (b)

Figure 14. (a) Effects of cooling and heating temperatures on the piston displacement S_p (b) the required external electric resistance $R_{e,out}$ to keep piston displacement at a typical allowable value of 8 mm.

6. Further discussions on thermoacoustic cogeneration systems

Besides the above-mentioned thermoacoustic trigeneration system, waste heat and LNG cold energy can also be recovered in a thermoacoustic cogeneration system. Combined cooling and electric power are usually required in summer. Cooling is conventionally produced from electricity which significantly increases the electricity peak demand. In winter, combined heating and electric power are often required. Heating is conventionally produced from burning fossil fuels that are undesirable for environmental reasons. A thermoacoustic cogeneration system may provide an effective way of tackling these issues. Figure 15 shows a thermoacoustic system for recovering waste heat and cold energy in three energy supply modes: a combined cooling, heating, and power (CCHP) model, a combined heating and power (CHP) model, and a combined cooling and power (CCP) model. Compared to the CCHP model, the CHP model uses the heat pump to simply produce heat power by pumping heat from

an ambient temperature and releasing it at a high temperature; the CCP model uses the heat pump to simply produce cooling power by pumping heat from a low temperature and releasing it at an ambient temperature. The overall exergy efficiencies for cogeneration systems are the same as Eq. (1), while Q_c is zero for the CHP system and Q_h is zero for the CCP system.

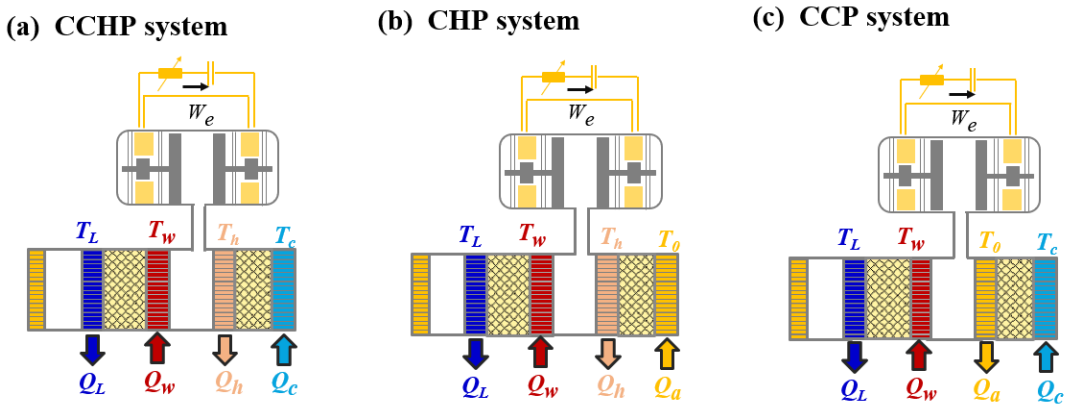


Figure 15. Schematic of the thermoacoustic multi-generation system in three energy supply modes of (a) combined cooling, heating and power (CCHP) (b) combined heating and power (CHP) (c) combined cooling and power (CCP), for waste heat and LNG cold energy recovery.

Figure 16 shows the thermodynamic performance of the thermoacoustic CHP system using waste heat and LNG cold energy. The results show that lower heating temperatures have a favorable effect on heating and electricity power but negatively impact on exergy efficiency. Figure 17 shows the thermodynamic performance of the thermoacoustic CCP system for waste heat and LNG cold energy recovery. Decreases in heating temperature have a positive effect on cooling and electric power but negligibly impact exergy efficiency.

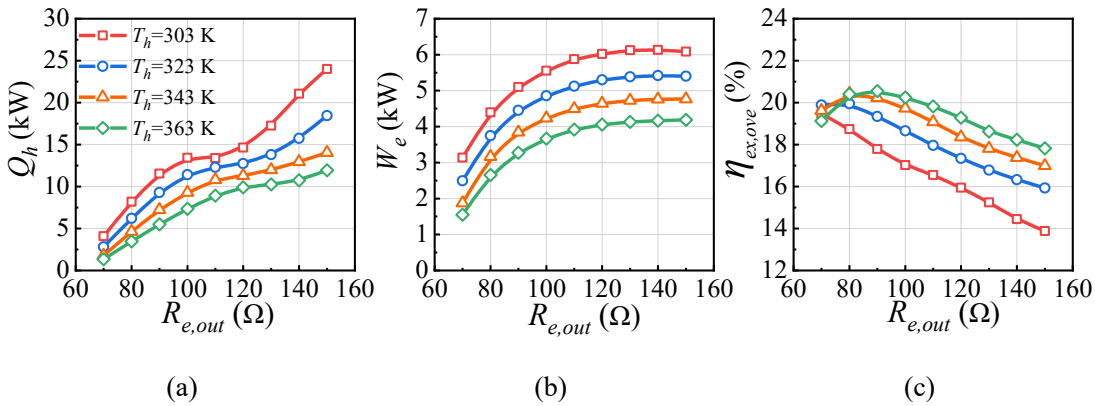
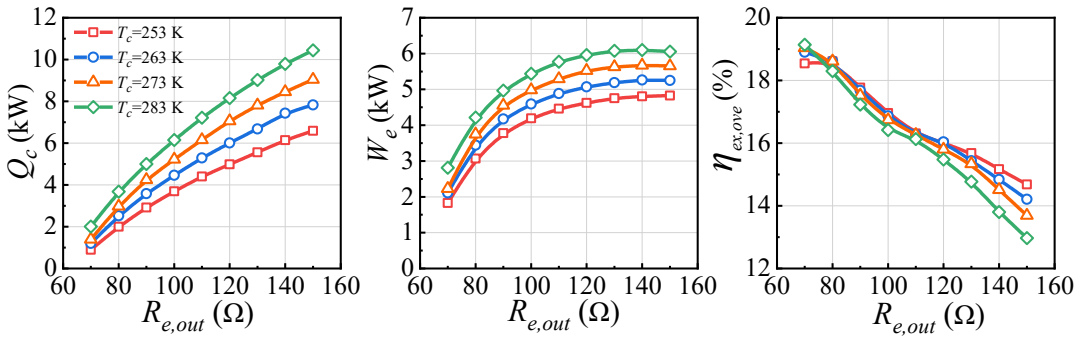


Figure 16. Thermodynamic performance of the thermoacoustic CHP system under different heating temperatures and electric resistances: (a) heating power; (b) electric power; (c) overall exergy efficiency. T_w is 523 K, T_L is 110 K, T_0 is 293 K, P_m is 4 MPa.



(a) (b) (c)

Figure 17. Thermodynamic performance of the thermoacoustic CCP system under different cooling temperatures and electric resistances: (a) cooling power; (b) electric power; (c) overall exergy efficiency. T_w is 523 K, T_L is 110 K, T_0 is 293 K, P_m is 4 MPa.

Table 6 presents summarized energy, cost saving, and emission savings for different energy supply models. The results demonstrate that the system displays similar energy performance, economic benefits, and environmental impact across the different models: 16.4 to 19.0 % for the exergy efficiency, 20200 to 21000 €/year for the annual energy cost saving, and 61100 to 64200 tCO_2/year for the CO_2 emission reduction. In the new framework of decentralized waste energy utilization, therefore, thermoacoustic technology offers an interesting alternative with its modular feature: only one thermoacoustic system is required, in conjunction with various energy sources, to operate in different energy supply modes for various applications.

Table 6. Performance comparison of the thermoacoustic system in the three typical energy supply models. T_w is 523 K, T_L is 110 K, T_h is 323 K, T_c is 283 K, T_0 is 293 K, P_m is 4 MPa.

	CHP system	CCP system	CCHP system
Pumped heating power (kW)	11.4	0	9.62
Cooling power (kW)	0	6.15	4.19
Electrical power (kW)	4.86	5.44	4.34
Exergy efficiency of the overall system (%)	18.7	16.4	19.0
Annual fuel energy saving (MWh/year)	187	116	175
Annual energy cost saving (€/year)	21000	20200	20700
Annual potential CO_2 emission reduction (tCO_2/year)	64200	61100	63100

7. Comparisons to similar technologies and future developments

7.1 Comparison with current thermoacoustic systems

The proposed strategy presents the first concept of jointly combining a thermoacoustic multi-generation system with the utilization of waste heat and cold energy. When compared to current waste-heat-recovery thermoacoustic systems, the proposed system enables much larger power amplification for the same heating temperatures or a much smaller heating temperature required for the same power amplification factor, due to the much larger temperature gradient in the regenerator. For example, one current thermoacoustic system achieves the regenerator power amplification factor of 1.32 at a heating temperature of 400 K [24], while the proposed system achieves the factor of 2.13 at the same heating temperature and a cooling temperature of 110 K. Furthermore, the novel concept of multiple power generation for combined cooling, heating and electricity is attractive for numerous applications.

At present, no similar thermoacoustic CCHP system exists for comparison in terms of system efficiency. However, the proposed advanced configuration enables high efficiency through the introduction of the following concepts. A looped multi-stage configuration is introduced to create a traveling-wave acoustic field leading to high efficiency. The efficiency of a looped multi-stage thermoacoustic cryocooler was shown to be increased by a factor of 1.5 relatively to single-stage system [47]. An efficiency gains of 28% also resulted from the elimination of the phase-shifter tube between the engine and the heat pump unit according to our recent simulations.

7.2 Comparison with exsiting systems for waste heat and LNG cold recovery

Most active thermodynamic cycles are based on Rankine or combined cycles, which are usually designed to using waste heat and recover cold energy to provide up to megawatts of power. Nevertheless, there are also sufficiently abundant distributed waste heat and cold energy sources in a much smaller scale. The proposed thermoacoustic system aims to harness such small-scale waste energy. Table 7 presents comparison results of the thermoacoustic system with current existing systems in terms of overall exergy efficiency. The results show that the proposed thermoacoustic system has comparable exergy to current large-scale systems. For example, Sun *et al.* showed the existing organic-Rankine-cycle based system achieved an exergy efficiency of 21.3% when using LNG cold energy and waste heat at 473 K [48]. Emadi *et al.* showed that an organic-Rankine-cycle based multiple-generation system achieved an exergy efficiency of 24.9%, by utilizing a geothermal heat source at 450 K and LNG cold energy [49]. Although some other existing systems show higher efficiencies than the thermoacoustic system [50, 51], the efficiency of the proposed system can be further improved by replacing the resonance tubes with mechanical pistons to reduce power

losses. In addition to the comparable efficiency, the thermoacoustic system also has the following benefits:

- a simple configuration by integrating multiple energy conversions into a single system,
- adaptability for use as a cogeneration or trigeneration system in various applications,
- no use of ozone-depleting or greenhouse gases,
- high reliability due to the lack of moving parts and low material requirements.

Table 7. Comparison of the thermoacoustic system with the other existing systems for waste heat and LNG cold recovery

Authors	System	Heat source and heat sink	Overall exergy efficiency
Kim *et al.* [52]	*Configuration*: an ammonia–water Rankine cycle and an LNG Rankine cycle *Output*: electricity	*Heat source*: low grade heat 473 K *Heat sink*: LNG fuel 111 K	17.2%
Sun *et al.* [48]	*Configuration*: Single-stage ORC *Output*: electricity	*Heat source*: low grade heat 473 K *Heat sink*: LNG fuel 111 K	21.3%
Tian *et al.* [53]	*Configuration*: Two-stage organic-Rankine-cycle (ORC) *Output*: electricity	*Heat source*: engine exhaust gas 503 K, jacket cooling water 368 K *Heat sink*: LNG fuel 111 K	23.3%
Emadi *et al.* [49]	*Configuration*: Two ORCs, an absorption refrigeration cycle, and a PEM electrolyzer *Output*: combined electricity, cooling power and hydrogen	*Heat source*: geothermal 450 K *Heat sink*: LNG fuel 111 K	24.9%
Liu *et al*. [50]	*Configuration*: three power cycles, a CO2 capture system, and a multistage expansion system *Output*: combined cooling, heating, and electricity	*Heat source*: low grade heat 669 K *Heat sink*: LNG fuel 111 K	38.3%
Li *et al*. [51]	*Configuration*: two power cycles and an absorption refrigeration cycle *Output*: combined cooling and electricity	*Heat source*: low grade heat 403 K *Heat sink*: LNG fuel 111 K	36.1%
This work	*Configuration*: thermoacoustic cycle *Output*: combined cooling, heating, and electricity	*Heat source*: low grade heat 500 K *Heat sink*: LNG fuel 130 K	24.1%

7.3 Future developments

Thermoacoustic technology offers great potential for utilizing small-scale waste energy in the future. The current analysis was done for systems of the order of kilowatts. However, this can be considered as a modular unit, which can be scaled up to hundreds of kilowatts without loss of performance, through incorporating different stages in the loop or enlarging system dimensions. One possible application is to utilize waste heat and LNG cold energy in LNG-fueled ship vessels. As shown in Figure 18, the primary power for the vessel is generated by burning natural gas to produce high-temperature combustion gases with sufficient energy to power the gas turbine. The thermoacoustic trigeneration system adopts the exhaust gas of the gas turbine as a heat source, and uses the cold energy of LNG as cold sources. Based on a similar working principle, the proposed system can also possibly be applied in distributed small-scale power plants to simultaneously recover low-temperature waste heat from power plants and utilise LNG cold energy from the LNG user terminal.

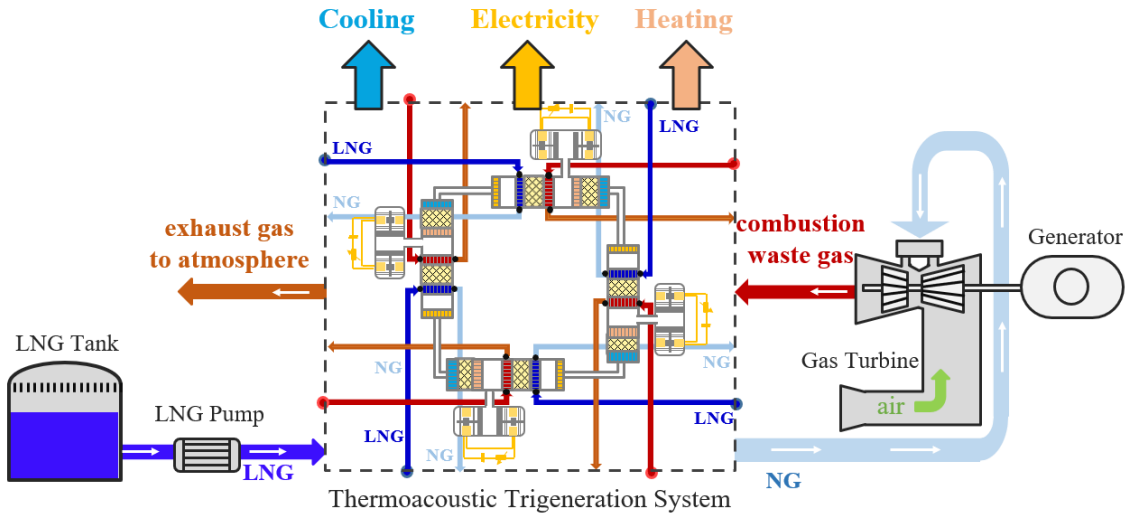


Figure 18. A case example of using the thermoacoustic trigeneration system for waste energy recovery in an LNG-fueled vessel.

While thermoacoustic technology shows promising prospects for future applications, it is still in early stages of development, and the demonstrators reported are not yet commercially viable. Currently, three disadvantages must be resolved in order for this technology to be successfully implemented in a variety of real-life applications. The first are the high investment costs, including relatively expensive component materials (e.g., large amounts of copper used in heat exchangers), high manufacturing costs (e.g., wire electrical discharge machining for gas channels) and initial operating charges (e.g., used for high-pressure helium), especially since most components are not available in the market at scale. The second challenge is to ensure a sufficient fatigue life, corresponding to at least several years of operation. The third difficulty is to ensure high efficiency, although the results in this work show that the efficiency of the proposed system is comparable to some existing systems. Resolving

these issues will help create a renewable energy conversion technology that has great potential for waste energy unitizations and will present a promising alternative to conventional technologies.

8. Conclusions

This paper introduces a first-ever thermoacoustic combined heating, cooling, and electricity system for simultaneously recovering waste heat and LNG cold energy. The proposed system integrates multiple energy conversion processes into one system and allows for small-scale waste energy utilization. Theoretical analyses have been carried out to investigate the characteristics of the system. The following conclusions can be drawn from the results:

(1) The optimized trigeneration system can achieve a combined cooling of 2.19 kW at 283 K, a heating power of 3.55 kW at 323 K and an electric power of 2.27 kW, with an overall exergy efficiency of 24.1 %, where LNG and waste heat temperatures are of 130 K and 500 K, respectively. By considering the economic and environmental aspects, this system allows for around 78.4 MWh of primary fuel energy consumption reduction, and can avoid around 30.6 tons of CO_2 emissions per year. These findings suggest that thermoacoustic technology has a significant potential to lower emissions and primary energy consumption.

(2) Load acoustic impedance is introduced to decouple the alternator and thermoacoustic unit for design optimization. By adjusting the load acoustic impedance, an optimum acoustic field can be achieved in the thermoacoustic unit in terms of the highest efficiency. When the optimal impedance for the inlet of the alternator is achieved, the maximum alternator efficiency is dependent on the piston diameter.

(3) The working parameters may deviate from the design values, which may lead to excessive piston displacement. Adjusting the external electric resistance is an effective way of avoiding this problem with a very short response. For instantaneous deviations in the operating temperatures, smaller electric resistance is required to limit larger displacement below the design value.

(4) Thermoacoustic technology can be used as a cogeneration or trigeneration system based on different energy supplies. The results indicate similar energetic, economic, and environmental potential across the different models. Such adaptability allows for the proposed system to be used in various applications.

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(5) Compared to current single-source driven thermoacoustic engines, the proposed combined-source engine achieves a much higher power amplification factor with useful efficiencies. Compared to other Rankine cycles for a similar purpose, the proposed system shows a comparable exergy efficiency with additional advantages of simple configuration, expected high reliability, and environmental friendliness.

This work suggests that thermoacoustic multi-generation systems may lead to opportunities for decentralized, small-scale waste energy recovery in e.g., LNG-fueled ship vessels or distributed power plants. Further efforts in system commercialization, cost reduction and efficiency enhancement are required to make thermoacoustic technology economically competitive with other mature energy alternatives.

Acknowledgments

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